



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/AGSR Division

FIRST SEMESTER 2026-27

Course Handout (Part – II)

Date: 11/08/2026

In addition to Part I (General handout for all course appended to timetable), this portion gives further specific details regarding the course.

Course No. : BITS F111
Course Title : Thermodynamics
Instructor-in-Charge : **Amit Jain**
Tutorial Instructors : Pradipta Chattopadhyay

1. Course Description

Concepts and laws of thermodynamics; thermodynamic properties; applications to closed and open systems; entropy and entropy generation; availability.

2. Scope and Objective

Thermodynamics deals with energy, matter, and the laws governing their interactions.

- To learn usefulness of laws of thermodynamics in the design of processes, devices, and systems involving effective utilization of energy and matter.
- To understand the fundamentals and concepts of the laws of thermodynamics as applied to control mass and control volume systems.
- To apply the fundamentals and concepts of the laws of thermodynamics to control mass and control volume systems and analyze them.
- To examine the performance of engineering devices in light of the laws of thermodynamics.

3. To define, understand, apply and analyze the concept of irreversibility and availability to thermodynamic systems and to evaluate second law efficiency of thermodynamics systems and devices.

4. Text book (TB):

- Fundamentals of Thermodynamics, 10^{ed} An Indian Adaptation, Authors Claus Borgnakke, Richard E. Sonntag, Souvik Bhattacharyya, Manoj Kumar Soni, ISBN: 9789354642210
<https://www.wileyindia.com/fundamentals-of-thermodynamics-10ed-an-indian-adaptation.html>

- **Booklet on Thermodynamic Tables, Figures & Charts**

5. Reference books (RB):

- Çengel Y.A. and Boles M.A., “Thermodynamics: an engineering approach”, Tata McGraw- Hill, 2019, 9th Ed.
- Moran M. J., Shapiro H. N., Boettner D. D and Baily M. B., “Principles of Engineering Thermodynamics”, Wiley, 2014 8th Ed



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6. Course Plan

Module 1: Basic Concepts and Definitions

Learning Objectives

- To define the terminologies and associated basic concepts of thermodynamics such as system, state, state postulate, equilibrium, process, and cycle.
- To review concepts of temperature, temperature scales, pressure, and absolute and gauge pressure.

Lecture No.	Topics to be covered	Text book Chap/Sec #
1-3	Introduction, thermodynamic systems and control volume, properties & state, process & cycle, force, energy, pressure, specific volume, zeroth law.	1

Module 2: Properties of Pure Substances

Learning Objectives

- To introduce the concept of a pure substance and discuss the physics of phase change processes.
- To illustrate the P-v, T-v, and P-T property diagrams and P-v-T surfaces of pure substances.
- To determine thermodynamic properties of pure substances and ideal gases from tables of property data and charts.

Lecture No.	Topics to be covered	Text book Chap/Sec #
4-7	Pure substance, Phase equilibrium, Independent properties, Tables of thermodynamic properties & their use	2
8-9	P-V-T behavior, Compressibility factor, Equation of state	2

Module 3: Energy Transfer

Learning Objectives

- To introduce and define the concept of heat and work for correct analysis of thermodynamics systems.
- To learn and determine different kinds of work like moving boundary work or PdV work commonly encountered in reciprocating devices such as automotive engines and compressors, electric work, etc.

Lecture No.	Text book Chap/Sec #	Text book Chap/Sec #
10-11	Definition of work and its identification, work done at the moving boundary Concept of heat, comparison of heat and work, Engineering Applications	3





Module 4: Energy Equation for Control Mass

Learning Objectives

- To understand and develop the general energy balance equation applied to Control Mass systems.
- To define the specific heat at constant volume and the specific heat at constant pressure.
- To relate the specific heats to the calculation of the changes in internal energy and enthalpy of ideal gases.

Lecture No.	Text book Chap/Sec #	Text book Chap/Sec #
12-13	First law for a cycle as well as for a change of state; internal energy & enthalpy	3
14-15	Specific heats; internal energy, enthalpy & specific heat of ideal gases; first law as a rate equation; problem analysis & solution technique, Engineering Applications	3

Module 5: Energy Equation for Control Volume

Learning Objectives

- To develop and apply the conservation of mass principle to various systems including steady- and unsteady-flow control volumes.
- To identify the energy carried by a fluid stream crossing a control surface as the sum of internal energy, flow work, kinetic energy, and potential energy of the fluid and to relate the combination of the internal energy and the flow work to the property enthalpy.
- To solve energy balance problems for common steady-flow devices such as nozzles, compressors, turbines, throttling valves, mixers, heaters, and heat exchangers.
- To apply the energy balance to general unsteady-flow processes with particular emphasis on the uniform-flow process as the model for commonly encountered charging and discharging processes.

Lecture No.	Text book Chap/Sec #	Lecture No.
16-19	Conservation of mass in control volume; first law for control volume; Steady state process; examples of SS processes	4
20-21	Transient processes; examples, Engineering Applications	4

Module 6: The Second Law of Thermodynamics

Learning Objectives

- To get an introduction to the second law of thermodynamics.
- To identify valid processes as those that satisfy both the laws of thermodynamics.





- To understand the concepts of thermal energy reservoirs, reversible and irreversible processes, heat engines, refrigerators, heat pumps, and the thermal efficiencies and coefficients of performance for reversible heat engines, heat pumps, and refrigerators.
- To understand the Kelvin–Planck and Clausius statements of the second law of thermodynamics, and apply them to cycles and cyclic devices.
- Apply the second law for developing the absolute thermodynamic temperature scale.
- To understand the Carnot cycle, and examine the Carnot principles.

Lecture No.	Topics to be covered	Text book Chap/Sec #
22-27	Limitations of first law & need for the second law; Reversible process; Heat engine, heat pump, refrigerator; Carnot cycle; Two prepositions regarding efficiency of Carnot cycle; Energy-conversion efficiency and COP, Kelvin-Planck & Clausius statements, Thermodynamic temperature scale, The ideal gas Carnot Cycle, Engineering Applications.	5

Module 7: Entropy

Learning Objectives

- Understand a new property called entropy for quantifying the second law effects.
- Analyze control mass systems by applying the entropy balance equation.
- Recognize processes that generate entropy and where the entropy is made.
- Calculate the entropy changes that take place during processes for pure substances, incompressible substances, and ideal gases.
- Apply the second law to general control mass systems.

Lecture No.	Topics to be covered	Text book Chap/Sec #
28-33	The inequality of Clausius, Concept of entropy; the Need of entropy definition of entropy; entropy of a pure substance; entropy change of a reversible & irreversible processes; principle of increase of entropy, thermodynamic property relation; problem analysis & solution techniques etc.	6

Module 8: Second-Law Analysis for a Control Volume

Learning Objectives

- Apply the second law to general control volumes.
- Analyze steady-state, single-flow devices, both reversible and irreversible.
- Extend the second law to transient processes.
- Apply the second law to multiple-flow devices with several inlets and outlets.
- Recognize when you can apply the Bernoulli equation or the expression for reversible shaft work.





- Application and analysis of an actual device using device efficiency.

Lecture No.	Topics to be covered	Text book Chap/Sec #
34-36	Second law for control volume; SS & Transient processes; SSSF process; principle of increase of entropy	7

Module 9: Irreversibility and Availability

Learning Objectives

- To define exergy, reversible work, and exergy destruction as a result of irreversibilities.
- To define the second-law efficiency.
- To develop the exergy balance relation apply it to closed systems and control volumes and analyze them.
- To evaluate second law efficiency of thermodynamics systems/devices.

Lecture No.	Topics to be covered	Text book Chap/Sec #
37-39	Available energy/Exergy, reversible work & irreversibility, non-reacting gas mix, Second law efficiency	8

Module 10: Thermodynamics Property Relations

Learning Objectives

- To develop fundamental relations between commonly encountered thermodynamic properties and express the
- properties that cannot be measured directly in terms of easily measurable properties like enthalpy, entropy etc.
- To develop the Maxwell, which form the basis for many thermodynamic relations
- To develop the Clapeyron equation and determine the enthalpy of vaporization from P, v, and T measurements alone.

Lecture No.	Text book Chap/Sec #	Text book Chap/Sec #
40	Clapeyron equation, Maxwell relations, Thermodynamic relation for enthalpy, internal energy, and entropy, expansively and compressibility factor, equation of state, generalized chart for enthalpy and entropy change, developing tables of property from experimental data	12





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Evaluation Scheme:

Evaluation Component (Syllabus)	Duration (minutes)	Weightage	Date and Time	Nature of Component
Tutorial tests (4 marks each)	15	20%	Announced 6 tests - best 5 will be considered for the final marks	Closed Book
Assessment linked to class participation	5-10	10%	Surprise test during lecture & tutorial class	Closed Book
Mid-semester Examination	90	30%	Refer AUGSD time-table	Opened Book
Comprehensive Examination	180	40%	Refer AUGSD time-table	Closed Book

Note: 1. Booklet on “Thermodynamic Tables, Figures & Charts”, as prescribed, will be allowed in the closed book tests also. However, it should not be defaced by writing any formula, equations, etc.

2. The evaluation scheme may change with prior notice.

3. The passing marks for this course will be determined based on one of the two criteria: 30% of the median of the post-comprehensive examination score or 20% of the average of the first three top scorers. The lower value between these two criteria will be considered as passing marks in this course.

Consultation Hour: To be announced by the respective instructor during the lecture/tutorial.

Notices: All notices concerning the course will be uploaded on the **Nalanda** portal only.

<https://nalanda.bits-pilani.ac.in/>

Attendance Policy:

Each student is expected to attend all classes and participate in all evaluation components.

Weightage of 10% is assigned for assessment linked to attendance and participation, towards this component a small surprise-quizzes will be conducted in the lecture and tutorial class. Best 5 quizzes linked to attendance and participation will be considered in the final score. This score will be considered for the students having minimum attendance of 50%. For the students having attendance less than 50%, zero marks will be assigned irrespective of quizzes attempted.



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Make-up:

- Make-up request for mid-semester examination is to be submitted 7 days in advance of the exam (or within 72 hrs in case of medical emergency). For comprehensive examination the request is to be made to AUGSD as per its guidelines.
- It is mandatory to submit the necessary supporting documents (such as: listed below) along with make-up request:
 - Letter of invitation/selection for a national/international event where the applicant is representing BITS Pilani.
 - Proof of hospitalization, medical certificate, prescription, medicine purchase bill, etc.
- The decision of granting make-up to the applicant is taken mutually by the team of course instructors and would be final in all such matters.
- ***There is no makeup for the tutorial as all the tutorial are announced and one tutorial in buffer.***

Prof. Amit Jain
Instructor-in-Charge
BITS F111

